SRINIVAS VENGALDAS

Software Engineer
Washington, DC | (202) 808-1999 | vengaldas02@gmail.com
https://www.linkedin.com/in/vengaldassrinivas/ | github.com/Srinivas-Vengaldas | vengaldas.vercel.app

EDUCATION

The George Washington University - *M.S., Information Systems and Technology* | *GPA: 3.68/4.0* — Expected Dec 2026
Mahindra University - *B.Tech., Computer Science and Engineering* — Jun 2024

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C/C++
Backend & APIs: Node.js, Fastify, FastAPI, Next.js, REST APIs, Idempotency, Rate Limiting, Caching, Authentication (JWT), Role-Based Access Control, LLM and RAG Integration, MCP Servers
Frontend: React, Vite, Tailwind CSS
Data, Infrastructure & Testing: PostgreSQL, Redis, Prisma, Docker, CI/CD (GitHub Actions), Grafana, Vitest, pytest, k6 load testing
CS Fundamentals: Data Structures, Algorithms, System Design, Object-Oriented Design, Operating Systems, Computer Architecture, Concurrency

EXPERIENCE

National University of Singapore - *Deep Learning Research Intern* — Jun 2022 – Jul 2022
TensorFlow, Keras, TensorFlow.js, React, GitHub Actions — srinivas-vengaldas.github.io/Autix

- Built Xception and VGG19 transfer-learning classifiers in TensorFlow on 2,940 face images, best 2022 run 86%: accuracy.
- Rebuilt and audited the work in 2026 as Autix: found 8 evaluation bugs, including swapped precision/recall and ROC AUC computed on rounded predictions, and retrained both models with fixed seeds on one shared 300-image test set.
- VGG19 reached 87.0% accuracy and 0.936 ROC AUC (Xception: 84.0%, 0.906); deployed both in the browser with TensorFlow.js so photos stay on-device.

PROJECTS

FraudGraph - Real-Time Payment Fraud Engine — github.com/Srinivas-Vengaldas/Fraud-Graph
TypeScript, Node.js, Fastify, Redis, PostgreSQL, Python, LightGBM, ONNX, React, k6

- Built a fraud engine with a TypeScript API (Fastify, Redis, PostgreSQL) that approves, reviews or declines transactions with reasons, replaying 590,540 real IEEE-CIS/Vesta transactions with exactly-once idempotency.
- Hand-wrote a Bloom-filter blocklist (1.05% measured false positives vs. 1% target), ring-buffer velocity windows about 1,000x faster than Redis sorted sets, and union-find identity rings with a supernode cap.
- At the same analyst workload, the LightGBM model caught 51.0% of fraud vs. 21.4% for rules alone.
- Verified idempotency under 10 parallel duplicate requests and fixed a stack overflow at 160K queued cases with a 300K-item regression test. Scoring: 1.78 ms p95, 952 req/s on a laptop.

SafeGate - LLM Guardrails Gateway — github.com/Srinivas-Vengaldas/SafeGate
Python, FastAPI, PyTorch, ONNX Runtime, Presidio, PostgreSQL, Redis, Docker, GitHub Actions, Gemini

- Designed a layered LLM gateway in FastAPI with YAML policies per app, where cheap rule rails run first, keeping full-policy overhead to 45/48 ms p50/p95 and about 1 ms on Redis cache hits.
- Added PostgreSQL decision logging, Redis rate limiting and Prometheus/Grafana monitoring, and fit the whole demo in about 390 MB on a 512 MB free host by memory-mapping the int8 model.
- Cut false positives on tricky safe prompts from 18.7% to 4.7% by adding hard-negative coding data to training.
- Added a guarded RAG pipeline on Gemini that screens documents at ingest and at retrieval and checks cited claims, raising detection of attacks hidden in retrieved documents from 43.5% to 96.5%.

DC Transit Pulse - Bus Reliability Data Lakehouse — dc-transit-pulse-eta.vercel.app
Python, PySpark, Databricks (Auto Loader, Delta Lake, Workflows), SQL, React, Vercel, GitHub Actions

- Built a 7-task scheduled Databricks pipeline (Auto Loader, PySpark, Delta Lake) that polls WMATA's live GTFS-Realtime bus feeds every 2 hours, ingesting 3.77M events (337K per full day) with 87 of 89 runs succeeding.
- Found one bus in four missed its schedule (75.4% on time over 45,592 departures), ranging 61% to 92% across 107 routes.
- Made every layer idempotent (watermarks, insert-only MERGE, checkpoints), removing 5.75% duplicate and late rows; reruns add 0 rows.

Smart DMV - AI Document Verification Portal (Graduate Capstone) — github.com/Srinivas-Vengaldas/Smart-DMV
Next.js, TypeScript, PostgreSQL (Prisma), Supabase, Google Cloud Vision, PaddleOCR, FastAPI, Gemini, Python

- Co-built a portal that moves DC DMV proof-of-residency checks from the counter to before the visit residents upload a lease, OCR extracts it, and an agent approves, routes to staff or flags each case with a stored reason.
- As project manager of a six-person team, owned the OCR and verification backend, including a self-hosted FastAPI and PaddleOCR service and a quality gate that rejects unreadable scans.
- Implemented decision rules that weight name (40%), address (40%) and ZIP (20%), blended with OCR confidence. Cases auto-approve at 0.85 or higher, become exceptions below 0.45 and go to staff review in between.